

Approximating Irrational NumbersAdditional Practice 7A.02.04 **ANSWER KEY**

1. Find the value of each. If there is no real value, write “no real value.”

$\sqrt{81}$ 9	$\sqrt{-81}$ no real value	$\sqrt[3]{64}$ 4	$\sqrt[3]{-64}$ -4
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Cube roots accept negatives; square roots do not.

2. Why is there no real value for $\sqrt{-81}$ when there is one for $\sqrt[3]{-64}$?

Squaring any real number gives a positive result or zero, so nothing squares to -81.

***Cubing a negative gives a negative, because three negative factors leave one negative behind.
So $(-4)^3 = -64$.***

The point needs both halves — why one is refused and why the other is not.

3. Select ALL of the numbers below that are irrational.

- | | |
|--|--|
| ☐ $\sqrt{36}$ | ☒ $\sqrt{40}$ |
| ☐ $\sqrt[3]{125}$ | ☒ $\sqrt[3]{50}$ |
| ☒ 2π | ☐ $\frac{3}{4}$ |

Three of the six. $\sqrt{36} = 6$ and $\sqrt[3]{125} = 5$ are whole numbers, and $3/4$ is a ratio of integers. 2π is π scaled, and scaling an irrational number by a NON-ZERO rational one leaves it irrational.

Questions 4 and 5: name the perfect squares or cubes first, then the root.

4. $\sqrt{95}$

- a. 95 lies between the perfect squares _____ and _____.

81 and 100

- b. So the root lies between the integers _____ and _____.

9 and 10

- c. The root is closer to _____.

10

$95 - 81 = 14$ and $100 - 95 = 5$, so 95 sits nearer 100.

5. $\sqrt[3]{110}$

- a. 110 lies between the perfect cubes _____ and _____.
64 and 125
- b. So the root lies between the integers _____ and _____.
4 and 5
- c. The root is closer to _____.
5

110 - 64 = 46 and 125 - 110 = 15, so 110 sits nearer 125. Both cubes are on the list from Lesson 3.

6. Approximate $\sqrt{95}$ to the nearest tenth. Fill in the table, then decide.

Try	Result	How far from 95
$9.7 \times \text{itself}$	94.09	0.91
$9.8 \times \text{itself}$	96.04	1.04

To the nearest tenth, the value is **9.7**

9.7 misses by 0.91 and 9.8 misses by 1.04. It is close, which is worth pointing out — the two whole numbers were not close at all.

7. Approximate $\sqrt[3]{110}$ to the nearest tenth. Fill in the table, then decide.

Try	Result	How far from 110
4.7 cubed	103.823	6.177
4.8 cubed	110.592	0.592

To the nearest tenth, the value is **4.8**

4.8 misses by only 0.592 while 4.7 misses by 6.177.

8. Use $\sqrt{95} \approx 9.747$ to approximate each expression to the nearest tenth.

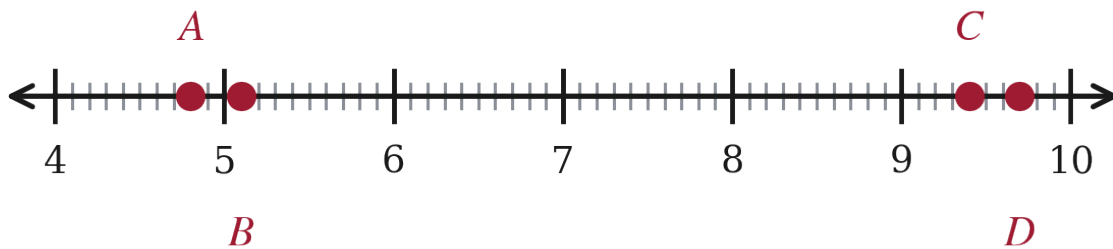
Expression	Approximate Value
$\sqrt{95} + 3$	12.7
$\frac{\sqrt{95}}{2}$	4.9
$2\sqrt{95}$	19.5

The sheet gives three decimal places on purpose. Rounding to 9.7 first and then doubling gives 19.4, which is a tenth short of the right answer. Question 11 is this idea asked directly.

9. Approximate the value of each expression to the nearest tenth.

	Expression	Approximate Value
A	$\sqrt[3]{110}$	4.8
B	$\pi + 2$	5.1
C	3π	9.4
D	$\sqrt{95}$	9.7

10. Plot and label A, B, C and D on the number line.



$A = 4.8$, $B = 5.1$, $C = 9.4$, $D = 9.7$. All four land on a tenth mark.

11. Marcus says that because $\sqrt{7} \approx 2.6$, the value of $3\sqrt{7}$ is 7.8. His calculator says 7.9. Who is right, and what went wrong?

The calculator is right. $3\sqrt{7} = 7.937\dots$, which is 7.9 to the nearest tenth.

Marcus rounded first and then multiplied. Rounding 2.6458 down to 2.6 lost 0.046, and tripling it lost 0.137.

Using $\sqrt{7} \approx 2.65$ gives $3 \times 2.65 = 7.95$, which rounds to 7.9 or 8.0 — better still, keep 2.646 and get 7.938.

Adding and subtracting are safe. Multiplying and dividing are not. Round last, or carry an extra place.